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ORGANIC LIGHT-EMITTING DIODE DISPLAY DEVICE INCLUDING A THIN FILM ENCAPSULATION LAYER

Abstract

A display device includes: a substrate; an organic light-emitting diode positioned above the substrate; and a thin film encapsulation layer disposed on the organic light-emitting diode. The thin film encapsulation layer includes: at least one inorganic layer; and at least one organic layer disposed on the at least one inorganic layer. The at least one organic layer has a refractive index of about 1.66 or greater.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/758,333 filed on Jun. 28, 2024, which is a continuation of U.S. patent application Ser. No. 17/739,314 filed on May 9, 2022, now U.S. Pat. No. 12,048,190, which is a continuation of U.S. patent application Ser. No. 16/810,107, filed on Mar. 5, 2020, now U.S. Pat. No. 11,329,257, issued on May 10, 2022, which is a continuation of U.S. patent application Ser. No. 15/704,218, filed on Sep. 14, 2017, now U.S. Pat. No. 10,608,207, issued on Mar. 31, 2020, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2016-0120123, filed on Sep. 20, 2016 in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] Exemplary embodiments of the present invention relate to a display device, and more particularly to an organic light-emitting diode (“OLED”) display device including a thin film encapsulation layer, the thin film encapsulation layer.

DISCUSSION OF RELATED ART

[0003] An organic light-emitting diode (“OLED”) display device may be a self-luminous display device, for example, that displays an image using an OLED emitting light. OLED display devices may have high quality characteristics such as a relatively low power consumption, a relatively high luminance, and a relatively high reaction speed.

[0004] The OLED display devices may have a multi-layered structure. The multi-layered structure may include an OLED. Due to such a structure, light emitted from the OLED may be extinguished when emitted to the outside. Thus, the OLED display device may have a relatively low luminous efficiency.

SUMMARY

[0005] Exemplary embodiments of the present invention provide a display device, and more particularly an organic light-emitting diode display device. The organic light-emitting diode display device may have a relatively low light emission deviation and relatively high light emission characteristics.

[0006] Exemplary embodiments of the present invention provide an organic light-emitting diode display device includes a substrate, an organic light-emitting diode, and a thin film encapsulation

layer. The organic light-emitting diode is positioned above the substrate. The thin film encapsulation layer is disposed on the organic light-emitting diode. The thin film encapsulation layer includes at least one inorganic layer and at least one organic layer. The at least one inorganic layer is disposed on the at least one inorganic layer. The at least one organic layer has a refractive index of about 1.66 or greater.

[0007] The at least one inorganic layer and the at least one organic layer may have a refractive index difference of about 0.06 or less.

[0008] The at least one organic layer may have a refractive index in a range of from about 1.66 to about 2.8.

[0009] The at least one inorganic layer may have a refractive index in a range of from about 1.6 to about 2.8.

[0010] The organic layer may be formed by polymerization of at least one monomer selected from: pentabromophenyl acrylate, 2-(9H-carbazol-9-yl) ethyl methacrylate, N-vinylcarbazole, bis(methacryloylthiophenyl) sulfide or zirconium acrylate.

[0011] The organic layer may include at least one selected from: poly(3,4-ethylenedioxythiophene) (PEDOT), 4,4'-bis [N-(3-methylphenyl)-N-phenylamino]biphenyl (TPD), 4,4',4''-tris [(3-methylphenyl)phenylamino] triphenylamine (m-MTDATA), 1,3,5-tris [N,N-bis(2-methylphenyl)-amino]benzene (0-MTDAB), 1,3,5-tris [N,N-bis(3-methylphenyl)-amino]benzene (m-MTDAB), 1,3,5-tris [N,N-bis(4-methylphenyl)amino]benzene (p-MTDAB), 4,4'-bis [N,N-bis(3-methylphenyl)-amino] diphenylmethane (BPPM), 4,4-dicarbazoyl-1,1'-biphenyl (CBP), 4,4',4''-tris(N-carbazole) triphenylamine (TCTA), 2,2',2''-(1,3,5-benzenetolyl) tris-[1-phenyl-1H-benzoimidazol] (TPBI) or 3-(4-biphenyl)-4-phenyl-5-t-butylphenyl-1,2,4-triazole (TAZ).

[0012] The at least one inorganic layer may include a first inorganic layer and a second inorganic layer. The at least one organic layer may include a first organic layer disposed between the first inorganic layer and the second inorganic layer.

[0013] The organic light-emitting diode may include a first electrode, an organic light-emitting layer, and a second electrode. The first electrode may be disposed on the substrate. The organic light-emitting layer may be disposed on the first electrode. The second electrode may be disposed on the organic light-emitting layer.

[0014] The organic light-emitting diode may include at least one of a hole injection layer and a hole transport layer disposed between the first electrode and the organic light emitting layer.

[0015] The organic light-emitting diode may include at least one of an electron transport layer and an electron injection layer disposed between the organic light emitting layer and the second electrode.

[0016] The organic light-emitting diode display device may further include a capping layer disposed between the organic light-emitting diode and the thin film encapsulation layer.

[0017] The organic light-emitting diode display device may further include a low refractive index layer disposed between the capping layer and the thin film encapsulation layer.

[0018] The low refractive index layer may have a refractive index in a range of from about 1.3 to about 1.4.

[0019] The low refractive index layer may include lithium fluoride (LiF).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] The above and other features of the present invention will become more apparent by describing in detail exemplary embodiments thereof, with reference to the accompanying drawings, in which:

[0021] FIG. 1 is a plan view illustrating an organic light-emitting diode (“OLED”) display device

according to an exemplary embodiment of the present invention;

[0022] FIG. 2 is a cross-sectional view taken along a line I-I' of FIG. 1 according to an exemplary embodiment of the present invention;

[0023] FIG. 3 is a cross-sectional view illustrating resonance in an OLED display device according to an exemplary embodiment of the present invention;

[0024] FIG. 4 is an emission spectrum graph of an OLED display device depending on a wavelength of emitted light according to an exemplary embodiment of the present invention;

[0025] FIGS. 5 and 6 are emission spectrum graphs of an OLED display device depending on a thickness of a first inorganic layer according to an exemplary embodiment of the present invention;

[0026] FIG. 7 is a cross-sectional view illustrating an OLED display device according to an exemplary embodiment of the present invention;

[0027] FIG. 8 is a cross-sectional view illustrating an OLED display device according to an exemplary embodiment of the present invention;

[0028] FIG. 9 is a cross-sectional view illustrating an OLED display device according to an exemplary embodiment of the present invention; and

[0029] FIG. 10 is a cross-sectional view illustrating an OLED display device according to an exemplary embodiment of the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0030] Exemplary embodiments of the present invention will be described below in more detail with reference to the accompanying drawings. In this regard, the exemplary embodiments may have different forms and should not be construed as being limited to the exemplary embodiments of the present invention described herein.

[0031] Like reference numerals may refer to like elements throughout the specification and drawings.

[0032] It will be understood that although the terms “first” and “second” may be used herein to describe various components, these components should not be limited by these terms.

[0033] Sizes of elements in the drawings may be exaggerated for clarity of description.

[0034] It will be understood that when a component, such as a layer, a film, a region, or a plate, is referred to as being “on” another component, the component can be directly on the other component or intervening components may be present.

[0035] Hereinafter, exemplary embodiments of the present invention will be described with reference to FIGS. 1 and 2.

[0036] FIG. 1 is a plan view illustrating an organic light-emitting diode (“OLED”) display device according to an exemplary embodiment of the present invention. FIG. 2 is a cross-sectional view taken along a line I-I' of FIG. 1 according to an exemplary embodiment of the present invention.

[0037] Referring to FIGS. 1 and 2, an OLED display device **101** according to an exemplary embodiment of the present invention may include a substrate **110**, a wiring unit **130**, an OLED **210**, and a thin film encapsulation layer **300**.

[0038] The substrate **110** may include an insulating material. The insulating material may include glass, quartz, ceramic, or plastic. However, exemplary embodiments of the present invention are not limited thereto. For example, the substrate **110** may include a metallic material such as stainless steel.

[0039] A buffer layer **120** may be disposed on the substrate **110**. The buffer layer **120** may include one or more layers selected from various inorganic layers or organic layers. The buffer layer **120** may substantially reduce or prevent undesirable elements, e.g., impurity elements or moisture, from penetrating into the wiring unit **130** or the OLED **210**. The buffer layer **120** may planarize a surface therebelow. The buffer layer **120** may be omitted.

[0040] The wiring unit **130** may be positioned above the buffer layer **120**. The wiring unit **130** may include a switching thin film transistor (“TFT”) **10**, a driving TFT **20** and a capacitor **80**. The wiring unit **130** may drive the OLED **210**. The OLED **210** may emit light, for example, according

to a driving signal received from the wiring unit **130** to display an image.

[0041] FIGS. **1** and **2** illustrate an active matrix-type organic light-emitting diode (AMOLED) display device **101** having a 2Tr-1Cap structure. For example, the 2Tr-1Cap structure may include two TFTs, e.g., the switching TFT **10** and the driving TFT **20**, and one capacitor **80** in each pixel; however, exemplary embodiments of the present invention are not limited thereto. For example, the OLED display device **101** may include three or more TFTs and two or more capacitors in each pixel. The OLED display device **101** may include additional wirings. Herein, the term “pixel” may refer to a smallest unit for displaying an image. The OLED display device **101** may display an image using a plurality of pixels.

[0042] Each pixel PX may include the switching TFT **10**, the driving TFT **20**, the capacitor **80**, and the OLED **210**. A gate line **151**, a data line **171**, and a common power line **172** may be disposed on the wiring unit **130**. The gate line **151** may extend in a first direction. The data line **171** and the common power line **172** may each be insulated from the gate line **151**. The data line **171** and the common power line **172** may intersect the gate line **151**. Each pixel PX may be defined by the gate line **151**, the data line **171** and the common power line **172** as a boundary; however, exemplary embodiments of the present invention are not limited thereto. The pixels PX may be defined by a pixel defining layer **190**.

[0043] The capacitor **80** may include a pair of capacitor plates **158** and **178**. An insulating interlayer **145** may be disposed between the pair of capacitor plates **158** and **178**. In an exemplary embodiment of the present invention, the insulating interlayer **145** may be a dielectric element. A capacitance of the capacitor **80** may be determined by electric charges accumulated in the capacitor **80** and a voltage across the pair of capacitor plates **158** and **178**.

[0044] The switching TFT **10** may include a switching semiconductor layer **131**, a switching gate electrode **152**, a switching source electrode **173**, and a switching drain electrode **174**. The driving TFT **20** may include a driving semiconductor layer **132**, a driving gate electrode **155**, a driving source electrode **176**, and a driving drain electrode **177**. A gate insulating layer **140** may be disposed on the buffer layer **110**. The gate insulating layer **140** may insulate the switching and driving semiconductor layers **131** and **132** and the switching and driving gate electrodes **152** and **155**.

[0045] The switching TFT **10** may be a switching element. The switching TFT **10** may select a pixel to perform light emission. The switching gate electrode **152** may be connected to the gate line **151**. The switching source electrode **173** may be connected to the data line **171**. The switching drain electrode **174** may be spaced apart from the switching source electrode **173**. The switching drain electrode **174** may be connected to at least one of the capacitor plates **158** or **178**, e.g., the capacitor plate **158**.

[0046] The driving TFT **20** may apply a driving power to a first electrode **211**. Thus, an organic light-emitting layer **212** of an OLED **210** in a selected pixel may emit light. The driving gate electrode **155** may be connected to the capacitor plate **158** connected to the switching drain electrode **174**. Each of the driving source electrode **176** and the other of the capacitor plates, e.g., the capacitor plate **178**, may be connected to the common power line **172**. The driving drain electrode **177** may be connected to the first electrode **211** of the OLED **210**, for example, through a contact hole.

[0047] The switching TFT **10** may be driven according to, for example, a gate voltage applied to the gate line **151**. The switching TFT **10** may transmit a data voltage applied to the data line **171** to the driving TFT **20**. A voltage equivalent to a difference between a common voltage applied to the driving TFT **20** from the common power line **172** and the data voltage transmitted by (or from) the switching TFT **10** may be stored in the capacitor **80**. A current corresponding to the voltage stored in the capacitor **80** may flow to the OLED **210** through the driving TFT **20**. Thus, the OLED **210** may emit light.

[0048] A planarization layer **146** may be disposed on the insulating interlayer **145**. The

planarization layer **146** may include an insulating material. The planarization layer **146** may protect the wiring unit **130**. The planarization layer **146** and the insulating interlayer **145** may include a same material as each other.

[0049] The OLED **210** may be disposed on the planarization layer **146**. The OLED **210** may include a first electrode **211**, the organic light-emitting layer **212**, and a second electrode **213**.

[0050] The organic light-emitting layer **212** may be disposed on the first electrode **211**. The second electrode **213** may be disposed on the organic light-emitting layer **212**. Holes and electrons may be injected into the organic light-emitting layer **212** from the first electrode **211** and the second electrode **213**, respectively. The holes and electrons may be combined in the organic light-emitting layer **212** and may form excitons. The OLED **210** may emit light when the excitons fall from an excited state to a ground state.

[0051] According to an exemplary embodiment of the present invention, the first electrode **211** may be an anode for injecting holes. The second electrode **213** may be a cathode for injecting electrons. However, exemplary embodiments of the present invention are not limited thereto. For example, the first electrode **211** may be a cathode, and the second electrode **213** may be an anode.

[0052] According to an exemplary embodiment of the present invention, the first electrode **211** may be a reflective film. The second electrode **213** may be a semi-transmissive film. Accordingly, a light generated in the organic light-emitting layer **212** may be emitted through the second electrode **213**. Thus, the OLED display device **101** according to an exemplary embodiment of the present invention may have a top emission type structure.

[0053] The first electrode **211** may have a structure in which, for example, a reflective film and a transparent conductive film are sequentially stacked. In such an exemplary embodiment of the present invention, the transparent conductive film of the first electrode **211** may be disposed between the reflective film and the organic light-emitting layer **212**.

[0054] The reflective film may include one or more metals selected from: magnesium (Mg), silver (Ag), gold (Au), calcium (Ca), lithium (Li), chromium (Cr), copper (Cu), or aluminum (Al). The reflective film may have a thickness of about 200 nm or greater.

[0055] The transparent conductive film may include a transparent conductive oxide (TCO). The TCO may include: indium tin oxide (ITO), indium zinc oxide (IZO), zinc oxide (ZnO), aluminum zinc oxide (AZO), or indium oxide (In.sub.2O.sub.3). The transparent conductive film may have a relatively high work function. Thus, hole injection through the first electrode **211** may be performed with relative ease.

[0056] The first electrode **211** may have a triple-layer structure in which a transparent conductive film, a reflective film and a transparent conductive film are sequentially stacked.

[0057] The second electrode **213** may include a semi-transmissive film. The semi-transmissive film may include one or more metals selected from magnesium (Mg), silver (Ag), gold (Au), calcium (Ca), lithium (Li), chromium (Cr), copper (Cu), or aluminum (Al). The semi-transmissive film may have a thickness of about 200 nm or less. As the thickness of the semi-transmissive film decreases, the transmittance of light may increase. As the thickness of the semi-transmissive film increases, the transmittance of light may decrease.

[0058] The organic light-emitting layer **212** may include a single molecule organic material or a polymer organic material. The organic light-emitting layer **212** may be formed by any known method using an organic light emitting material.

[0059] The pixel defining layer **190** may have an opening. The opening of the pixel defining layer **190** may expose a portion of the first electrode **211**. The organic light-emitting layer **212** and the second electrode **213** may be sequentially stacked on the first electrode **211** exposed by the opening. The second electrode **213** may be disposed on the pixel defining layer **190** and the organic light emitting layer **212**. The OLED **210** may emit light, for example, from the organic light-emitting layer **212** at the opening of the pixel defining layer **190**. Thus, the pixel defining layer **190** may define a light emission area.

[0060] The thin film encapsulation layer **300** may be disposed on the second electrode **213**, for example, to protect the OLED **210**. The thin film encapsulation layer **300** may reduce or prevent outside air such as moisture or oxygen from permeating into the OLED **210**.

[0061] For example, the thin film encapsulation layer **300** may include at least one inorganic layer **311** or **312** and at least one organic layer **321**. The at least one inorganic layer **311** or **312** and the at least one organic layer **321** may be alternately disposed.

[0062] According to an exemplary embodiment of the present invention, the at least one organic layer **321** may have a refractive index of about 1.66 or greater. For example, the at least one organic layer **321** may have a refractive index ranging from about 1.66 to about 2.8. The at least one inorganic layer **311** and **312** may have a refractive index ranging from about 1.6 to about 2.8.

[0063] The at least one inorganic layer **311** and **312** and the at least one organic layer **321** may have a refractive index difference of about 0.06 or less. When the refractive index difference is relatively small, light reflection at an interfacial surface between the at least one inorganic layer **311** and **312** and the at least one organic layer **321** may be reduced or prevented. Accordingly, optical resonance in the OLED display device **101** may be reduced or prevented, and a luminance deviation due to optical resonance may be reduced or prevented. The optical resonance will be described in more detail below.

[0064] Referring to FIG. 2, the thin film encapsulation layer **300** may include two inorganic layers **311** and **312** and one organic layer **321**. For example, the at least one inorganic layer may include a first inorganic layer **311** and a second inorganic layer **312**. The at least one organic layer may include a first organic layer **321**. The first inorganic layer **321** may be disposed between the first inorganic layer **311** and the second inorganic layer **312**. The at least one inorganic layer **311** and **312** may be referred to as an inorganic layer **311** and **312**. The at least one organic layer **321** may be referred to as an organic layer **321**. However, exemplary embodiments of the present invention are not limited to the structure described with reference to FIG. 2.

[0065] According to an exemplary embodiment of the present invention, the inorganic layer **311** and **312** may have a thickness in a range of from about 100 nm to about 1,000 nm. The inorganic layer **311** and **312** may include at least one of silver, metal oxide, metal oxynitride, silicon oxide, silicon nitride, or silicon oxynitride. The inorganic layer **311** and **312** may have substantially a same refractive index as each other. The inorganic layers **311** and **312** may have substantially a same thickness as each other. The inorganic layers **311** and **312** may include a same material as each other.

[0066] For example, the inorganic layer **311** and **312** may include one or more inorganic materials of: Al.sub.xO.sub.y, TiO.sub.x, ZrO.sub.x, SiO.sub.x, AlO.sub.xN.sub.y, Al.sub.xN.sub.y, SiO.sub.xN.sub.y, Si.sub.xN.sub.y, ZnO.sub.x or Ta.sub.xO.sub.y. For example, the inorganic layer **311** and **312** may include one or more inorganic materials of: Al.sub.2O.sub.3, TiO.sub.2, ZrO, SiO.sub.2, AlON, AlN, SION, Si.sub.3N.sub.4, ZnO or Ta.sub.2O.sub.5. The inorganic layer **311** and **312** may be formed through methods such as a chemical vapor deposition (CVD) method or an atomic layer deposition (ALD) method. However, exemplary embodiments of the present invention are not limited thereto. For example, the inorganic layer **311** and **312** may be formed through various methods.

[0067] The organic layer **321** may have a thickness in a range of from about 100 nm to about 8,000 nm. According to an exemplary embodiment of the present invention, the organic layer **321** may include an organic material having a relatively high refractive index. For example, the organic layer **321** may include a polymer material having a relatively high refractive index.

[0068] According to an exemplary embodiment of the present invention, the organic layer **321** may be formed by polymerization of at least one monomer selected from the following Chemical Formulae 1, 2, 3, 4 and 5.

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[0069] Thus, the organic layer **321** may be formed by polymerization of at least one monomer

selected from: pentabromophenyl acrylate, 2-(9H-carbazol-9-yl) ethyl methacrylate, N-vinylcarbazole, bis(methacryloylthiophenyl) sulfide or zirconium acrylate.

[0070] For example, the organic layer **321** may be formed by depositing, on the inorganic layer **311** or **312**, at least one monomer selected from: pentabromophenyl acrylate, 2-(9H-carbazol-9-yl) ethyl methacrylate, N-vinylcarbazole, bis(methacryloylthiophenyl) sulfide or zirconium acrylate.

[0071] The organic layer **321** including pentabromophenyl acrylate represented by Chemical Formula 1 may have a refractive index of about 1.7.

[0072] The organic layer **321** including 2-(9H-carbazol-9-yl) ethyl methacrylate represented by Chemical Formula 2 may have a refractive index of about 1.69.

[0073] The organic layer **321** including N-vinylcarbazole represented by Chemical Formula 3 may have a refractive index of about 1.68.

[0074] The organic layer **321** including bis(methacryloylthiophenyl) sulfide represented by Chemical Formula 4 may have a refractive index of about 1.66.

[0075] The organic layer **321** including zirconium acrylate represented by Chemical Formula 5 may have a refractive index of about 1.66. The refractive index of the organic layer **321** may be adjusted, for example, by adjusting the content of zirconium (Zr).

[0076] The organic layer **321** may include an acrylic resin, an epoxy resin, polyimide or polyethylene, which are polymer-based materials. For example, according to an exemplary embodiment of the present invention, the organic layer **321** may include at least one selected from: poly(3,4-ethylenedioxythiophene) (PEDOT), 4,4'-bis [N-(3-methylphenyl)-N-phenylamino]biphenyl (TPD), 4,4',4''-tris [(3-methylphenyl)phenylamino] triphenylamine (m-MTDATA), 1,3,5-tris [N,N-bis(2-methylphenyl)-amino]benzene (o-MTDAB), 1,3,5-tris [N,N-bis(3-methylphenyl)-amino]benzene (m-MTDAB), 1,3,5-tris [N,N-bis(4-methylphenyl)amino]benzene (p-MTDAB), 4,4'-bis [N,N-bis(3-methylphenyl)-amino] diphenylmethane (BPPM), 4,4-dicarbazolyl-1,1'-biphenyl (CBP), 4,4',4''-tris(N-carbazole) triphenylamine (TCTA), 2,2',2''-(1,3,5-benzenetolyl) tris-[1-phenyl-1H-benzoimidazol] (TPBI) or 3-(4-biphenyl)-4-phenyl-5-t-butylphenyl-1,2,4-triazole (TAZ).

[0077] The organic layer **321** may be formed, for example, through a thermal deposition process. The thermal deposition process may be performed within a temperature range that does not damage the OLED **210**. However, exemplary embodiments of the present invention are not limited thereto. For example, the organic layer **321** may be formed through various methods.

[0078] The inorganic layer **311** or **312** may have a relatively high density of thin film. Thus, the inorganic layer **311** or **312** may substantially reduce or prevent permeation of, for example, moisture or oxygen. Permeation of moisture and oxygen into the OLED **210** may be reduced or prevented by the inorganic layer **311** or **312**.

[0079] Moisture and oxygen that have passed through the inorganic layer **311** or **312** may be substantially blocked by the organic layer **321**. The organic layer **321** may have a relatively low moisture-infiltration preventing efficacy, as compared to the inorganic layer **311** and **312**. However, the organic layer **321** may be a buffer layer. Thus, the organic layer **321** may reduce stress among respective ones of the inorganic layer **311** and **312**. The organic layer **321** may also have a moisture-infiltration preventing function. Further, since the organic layer **321** may have planarization characteristics, an uppermost surface of the thin film encapsulation layer **300** may be planarized by the organic layer **321**.

[0080] The thin film encapsulation layer **300** may have a thickness of about 50 μm or less, e.g., about 10 μm or less. Accordingly, the OLED display device **101** may have a relatively small thickness. Thus, the OLED display device **101** may have a relatively high flexibility.

[0081] FIG. 3 is a cross-sectional view illustrating resonance in an OLED display device according to an exemplary embodiment of the present invention. Referring to FIG. 3, an intermediate layer **330** may be disposed between the OLED **210** and the thin film encapsulation layer **300**. The intermediate layer **330** may include a light-transmitting material. The intermediate layer **330** may

have a single-layered structure. Alternatively, the intermediate layer **330** may have a multi-layered structure in which a plurality of layers are stacked. The intermediate layer **330** may be omitted. [0082] The OLED display device **S1** may have a multi-layered structure. A light emitted from the organic light-emitting layer **212** may be emitted through the multi-layered structure of the OLED display device **S1**.

[0083] When a refractive index difference between the organic layer **321** and the inorganic layer **311** or **312** included in the thin film encapsulation layer **300** is relatively large, light may be reflected at an interfacial surface between the organic layer **321** and the inorganic layer **311** or **312**. Accordingly, optical resonance may be induced by the light reflection.

[0084] Optical resonance may be referred to herein as “resonance.” The resonance will be described with respect to the organic layer **321** and the first inorganic layer **311**, which may be one of the inorganic layers **311** and **312** positioned closest to the OLED **210**.

[0085] For example, referring to FIG. 3, the first inorganic layer **311** may have a refractive index of about 1.75. The organic layer **321** may have a refractive index of about 1.5.

[0086] When the first inorganic layer **311** has a refractive index of about 1.75 and the organic layer **321** has a refractive index of about 1.5, a refractive index difference between the first inorganic layer **311** and the organic layer **321** may be about 0.25. In such an exemplary embodiment of the present invention, an interfacial surface **311a** between the first inorganic layer **311** and the organic layer **321** may be a reflective surface. The first electrode **211** of the top emission type OLED display device may be a reflective electrode. Accordingly, light may be reflected off a surface **211a** of the first electrode **211**.

[0087] When light repeats reflection between two reflective surfaces, optical resonance may occur. When optical resonance occurs, light energy may increase. The light having the increased energy may relatively readily pass through the multi-layered stacked structure of the OLED display device **S1** to be emitted outwards. A structure allowing light resonance between two reflective layers may be referred to as a resonance structure. A distance between the two reflective layers between which resonance occurs may be referred to as a resonance distance. The resonance distance may vary, for example, based on the wavelength of light.

[0088] For example, when a wavelength of light emitted from the organic light-emitting layer **212** is “ 2π ” and a distance between the surface **211a** of the first electrode **211** and the interfacial surface **311a** between the first inorganic layer **311** and the organic layer **321** is “ d_1 ,” light resonance may occur when the following Formula 1 is satisfied.

$$d_1 = m_1 \cdot \lambda_1 \quad [\text{Formula 1}]$$

[0089] In Formula 1, “ m_1 ” is a natural number.

[0090] According to an exemplary embodiment of the present invention, the first inorganic layer **311** may be formed through methods such as a chemical vapor deposition (CVD) method or an atomic layer deposition (ALD) method. In such an exemplary embodiment of the present invention, due to process errors, a layer thickness may vary. Thus, a layer thickness may vary, for example, based on the position in a single first inorganic layer **311**.

[0091] When the layer thickness of the first inorganic layer **311** varies, a distance between the surface **211a** of the first electrode **211** and the interfacial surface **311a** between the first inorganic layer **311** and the organic layer **321** may vary. Thus, the wavelength of light resonated between the surface **211a** of the first electrode **211** and the interfacial surface **311a** between the first inorganic layer **311** and the organic layer **321** may vary.

[0092] FIG. 4 is an emission spectrum graph of an OLED display device according to the wavelength according to an exemplary embodiment of the present invention. In FIG. 4, C1 (represented by a dotted line) may refer to a relative luminous intensity based on the wavelength of light when the thickness of the first inorganic layer **311** is about 1 μm . C2 (represented by a bold line) may refer to a relative luminous intensity based on the wavelength of light when the thickness

of the first inorganic layer **311** is about 1.05 μm . Referring to FIG. 4, although the thickness of the first inorganic layer **311** may be changed by about 0.05 μm (i.e., about 50 nm), the luminous intensity based on the wavelength of light may vary.

[0093] The OLED display device may include a plurality of pixels. Due to process errors, the first inorganic layer **311** on the OLED **210** may have different thicknesses corresponding to respective pixels. For example, when the thickness of the first inorganic layer **311** on red OLEDs R is not uniform, luminous intensity of a red color may vary corresponding to each pixel, which may be similar to a green pixel and a blue pixel.

[0094] FIG. 5 is an emission spectrum graph of an OLED display device according to the thickness of the first inorganic layer according to an exemplary embodiment of the present invention. Referring to FIG. 5, as the thickness of the first inorganic layer **311** varies, luminous intensities of a red color R, a green color G, and a blue color B may vary.

[0095] As such, in the case where the interfacial surface **311a** between the first inorganic layer **311** and the organic layer **321** is a reflective layer, when the thickness of the first inorganic layer **311** varies, the wavelength of light resonated between the surface **211a** of the first electrode **211** and the interfacial surface **311a** between the first inorganic layer **311** and the organic layer **321** may vary. In such an exemplary embodiment of the present invention, a limit in forming a uniform thickness of the first inorganic layer **311** in a process of manufacturing the first inorganic layer **311** and the thickness of the first inorganic layer **311** may vary based on the position due to process errors. Accordingly, pixels representing a same color may exhibit different luminous intensities based on the position.

[0096] According to an exemplary embodiment of the present invention, at least the organic layer **321** may have a refractive index of about 1.66 or more. For example, the organic layer **321** may have a refractive index in a range of from about 1.66 to about 2.8. Accordingly, a refractive index difference between the inorganic layer **311** or **312** and the organic layer **321** may be reduced to about 0.06 or less. Thus, light reflection at the interfacial surface between the inorganic layer **311** and **312** and the organic layer **321** may be reduced or prevented.

[0097] According to an exemplary embodiment of the present invention, a refractive index difference between the first inorganic layer **311** and the organic layer **321** may be relatively small since the organic layer **321** may include a relatively high refractive index material and may have a relatively large refractive index of about 1.66 or more. Thus, the first inorganic layer **311** and the organic layer **321** may have a refractive index difference of about 0.06 or less. The first inorganic layer **311** and the organic layer **321** may have substantially a same refractive index as each other (e.g., the refractive index difference=0). Thus, light might not be reflected between the first inorganic layer **311** and the organic layer **321**. Accordingly, even if the thickness of the first inorganic layer **311** varies depending on the position due to process errors, there may be substantially no deviation in luminous efficiency due to resonance of light. Even if there is a deviation in the thickness of the first inorganic layer **311**, there is no deviation in the luminous intensity.

[0098] Accordingly, the OLED display device **101** according to an exemplary embodiment of the present invention may have a substantially uniform luminous efficiency for each pixel or position.

[0099] FIG. 6 is an emission spectrum graph of an OLED display device according to the thickness of the first inorganic layer according to an exemplary embodiment of the present invention. The OLED display device may refer to the OLED display device **101** according to an exemplary embodiment of the present invention.

[0100] Referring to FIG. 6, even if the thickness of the first inorganic layer **311** varies, luminous intensities of a red pixel R, a green pixel G and a blue pixel B might not substantially vary. Thus, there may be no substantial deviation in luminous intensity depending on the thickness of the first inorganic layer **311**.

[0101] An exemplary embodiment of the present invention will be described in more detail below

with reference to FIG. 7.

[0102] FIG. 7 is a cross-sectional view illustrating an OLED display device according to an exemplary embodiment of the present invention. Descriptions of the above-described components may be omitted below.

[0103] The OLED display device **102** according to an exemplary embodiment of the present invention may include a first light emission auxiliary layer **215**. The first light emission auxiliary layer **215** may be disposed between a first electrode **211** and an organic light-emitting layer **212**. The first light emission auxiliary layer **215** may include at least one of a hole injection layer HIL and a hole transport layer HTL. The first light emission auxiliary layer **215** may include the hole injection layer and the hole transport layer.

[0104] The OLED display device **102** according to an exemplary embodiment of the present invention may include a second light emission auxiliary layer **217**. The second light emission auxiliary layer **217** may be disposed between the organic light emitting layer **212** and a second electrode **213**. The second light emission auxiliary layer **217** may include at least one of an electron transport layer ETL and an electron injection layer EIL. The second light emission auxiliary layer **217** may include the electron injection layer EIL and the electron transport layer ETL.

[0105] The first light emission auxiliary layer **215** and the second light emission auxiliary layer **217** may each extend between a pixel defining layer **190** and the second electrode **213**.

[0106] The organic light-emitting layer **212**, the hole injection layer, the hole transport layer, the electron transport layer and the electron injection layer may also be referred to as an organic layer. The organic layer may include a relatively low molecular weight organic material or a relatively high molecular weight organic material.

[0107] The OLED display device **102** according to an exemplary embodiment of the present invention may include the first light emission auxiliary layer **215** or the second light emission auxiliary layer **217**. Alternatively, the OLED display device **102** may include the first light emission auxiliary layer **215** and the second light emission auxiliary layer **217**.

[0108] An exemplary embodiment of the present invention will be described in more detail below with reference to FIG. 8.

[0109] FIG. 8 is a cross-sectional view illustrating an OLED display device according to an exemplary embodiment of the present invention.

[0110] The OLED display device **103** according to an exemplary embodiment of the present invention may include a capping layer **230**. The capping layer **230** may be disposed between an OLED **210** and a thin film encapsulation layer **300**. Referring to FIG. 8, the capping layer **230** may be disposed between a second electrode **213** and a first inorganic layer **311**.

[0111] The capping layer **230** may have optical transparency. The capping layer **230** may protect the OLED **210**. The capping layer **230** may provide a light emitted from an organic light-emitting layer **212** to be emitted to the outside.

[0112] The capping layer **230** may include at least one of an inorganic material and an organic material having light transmittance. Thus, the capping layer **230** may include an inorganic layer or an organic layer. Alternatively, the capping layer **230** may include an organic layer including inorganic particles.

[0113] The capping layer **230** may include two or more materials having different refractive indices. For example, the capping layer **230** may be formed by using a mixture of a relatively high refractive index material and a relatively low refractive index material. The relatively high refractive index material and the relatively low refractive index material may be an organic material or an inorganic material, respectively.

[0114] The capping layer **230** may have a thickness in a range of from about 30 nm to about 300 nm. Alternatively, the capping layer **230** may have a thickness of about 300 nm or more. As the thickness of the capping layer **230** increases, protection of the OLED **210** may increase. However, if the capping layer **230** is relatively thick, a thickness of the OLED display device **103** may be

increased.

[0115] The capping layer **230** may be prepared by various methods. For example, the capping layer **230** may be formed by a deposition process. In the deposition process, a relatively high refractive index material and a relatively low refractive index material may be used together. A deposition amount or a deposition ratio of the relatively high refractive index material and the relatively low refractive index material may be adjusted, for example, to adjust a refractive index of the capping layer **230**. The capping layer **230** may have a refractive index in a range of from about 1.3 to about 1.4. For example, the capping layer **230** may have a refractive index in a range of from about 1.35 to about 1.38. For example, the capping layer **230** may have a refractive index of about 1.36.

[0116] FIG. **9** is a cross-sectional view illustrating an OLED display device according to an exemplary embodiment of the present invention.

[0117] The OLED display device **104** according to an exemplary embodiment of the present invention may include a capping layer **230** and a low refractive index layer **240**. The capping layer **230** and the low refractive index layer **240** may be sequentially disposed between an OLED **210** and a first inorganic layer **311**.

[0118] The low refractive index layer **240** may have a refractive index in a range of from about 1.3 to about 1.4. The low refractive index layer **240** may have a refractive index, for example, in a range of about 1.35 to about 1.39. For example, the low refractive index layer **240** may have a refractive index of about 1.37. The low refractive index layer **240** may provide a light emitted from the organic light-emitting layer **212** to be emitted to the outside.

[0119] The low refractive index layer **240** may include lithium fluoride (LiF).

[0120] FIG. **10** is a cross-sectional view illustrating an OLED display device according to an exemplary embodiment of the present invention.

[0121] According to an exemplary embodiment of the present invention, a thin film encapsulation layer **300** may include a plurality of inorganic layers **311**, **312** and **313** and a plurality of organic layers **321** and **322**.

[0122] For example, the thin film encapsulation layer **300** described with reference to FIG. **10** includes three inorganic layers **311**, **312** and **313** and two organic layers **321** and **322**. The two organic layers **321** and **322** may be alternately arranged with the three inorganic layers **311**, **312** and **313**.

[0123] The three inorganic layers **311**, **312** and **313** may be a first inorganic layer **311**, a second inorganic layer **312** and a third inorganic layer **313**, respectively. The two organic layers **321** and **322** may be named as a first organic layer **321** and a second organic layer **322**, respectively.

[0124] In one or more exemplary embodiments of the present invention, the thin film encapsulation layer of the OLED display device may include an organic layer having a relatively high refractive index. Accordingly, a refractive index difference between the organic layer and the inorganic layer included in the thin film encapsulation layer may be reduced, and the light emission characteristics of the OLED display device may increase.

[0125] While the present invention has been illustrated and described with reference to the exemplary embodiments thereof, it will be apparent to those of ordinary skill in the art that various changes in form and detail may be made thereto without departing from the spirit and scope of the present invention.

Claims

1. A display device, comprising: a substrate; a light-emitting diode positioned above the substrate; and a thin film encapsulation layer disposed on the light-emitting diode; and wherein the light-emitting diode comprises: a first electrode disposed on the substrate; a light-emitting layer disposed on the first electrode; and a first light emission auxiliary layer disposed between the first electrode and the light-emitting layer, wherein the thin film encapsulation layer comprises: an inorganic layer

- disposed on the light-emitting diode; and an organic layer disposed on the inorganic layer, wherein the organic layer has a refractive index in a range from about 1.66 to about 2.8, the inorganic layer has a refractive index in a range from about 1.6 to about 2.8, and the first light emission auxiliary layer is disposed between the first electrode and the light-emitting layer.
2. The display device of claim 1, wherein the first light emission auxiliary layer comprises at least one of a hole injection layer and a hole transport layer.
3. The display device of claim 1, further comprising: a second electrode disposed on the light-emitting layer, and a second light emission auxiliary layer disposed between the light-emitting layer and the second electrode.
4. The display device of claim 3, wherein the second light emission auxiliary layer comprises at least one of an electron transport layer and an electron injection layer.
5. The display device of claim 1, further comprising: a first layer disposed between the light-emitting diode and the thin film encapsulation layer, wherein a refractive index of the first layer is less than the refractive index of the organic layer.
6. The display device of claim 5, wherein the refractive index of the first layer is less than the refractive index of the inorganic layer.
7. The display device of claim 1, further comprising: a second layer disposed between the first layer and the thin film encapsulation layer, wherein a refractive index of the second layer is less than the refractive index of the inorganic layer.
8. A display device, comprising: a substrate; a light-emitting diode disposed on the substrate; a thin film encapsulation layer disposed on the light-emitting diode; a first layer disposed between the light-emitting diode and the thin film encapsulation layer; and a second layer disposed between the first layer and the thin film encapsulation layer, wherein the thin film encapsulation layer comprises: an inorganic layer disposed on the light-emitting diode; and an organic layer disposed on the inorganic layer, wherein a refractive index of the first layer is less than a refractive index of the organic layer, and wherein a refractive index of the second layer is less than the refractive index of the organic layer.
9. The display device of claim 8, wherein the refractive index of the first layer is in a range of from about 1.3 to about 1.4.
10. The display device of claim 8, wherein the refractive index of the organic layer is in a range of from about 1.66 to about 2.8.
11. The display device of claim 10, wherein a refractive index difference between the inorganic layer and the organic layer is about 0.06 or less.
12. The display device of claim 8, wherein the refractive index of the first layer is less than a refractive index of the inorganic layer.
13. The display device of claim 12, wherein the refractive index of the inorganic layer is in a range from about 1.6 to about 2.8.
14. The display device of claim 12, wherein the refractive index of the second layer is less than the refractive index of the inorganic layer.
15. The display device of claim 1, wherein the second layer comprises lithium fluoride (LiF), and wherein the refractive index of the second layer is in a range of from about 1.3 to about 1.4.
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